

and relevant information to farmers as well as private sector organisations working in the agriculture ecosystem to create a more informed and precise demand-supply chain. IDEA focuses on the creation of digital readiness, open data sources, OneAPI, and innovation to reduce the cost of smart farming,” she explains.

In December 2022, the Department of Agriculture and Farmers Welfare announced that it has commenced efforts to create Agristack, which is a collection of technologies and digital databases focusing on India’s farmers and the agricultural sector. The platform will serve as a farmer interface for ‘smart and well-organised agriculture’ and help tackle issues such as poor access to credit and wastage in the agricultural supply chain. The department entered into memorandums of understanding (MoUs) with leading technology companies, including Microsoft India, Amazon Web Services India, ITC, Cisco, and Ninjabart, to develop proof-of-concept for different focus areas or layers of the platform.

Jalan remarks that overcoming infrastructure challenges is crucial for our country to make it more Atma Nirbhar. “Adequate infrastructure improves farm productivity and reduces costs. There are several efforts already taken by our government to improve it. For example, the plan to set up a digital public infrastructure for agriculture is a welcome step, which ensures enhanced connectivity, and streamlines the supply chain, which in turn improves production. Another recent announcement was to set up 100 labs in engineering institutions, for developing applications using 5G services, which will further accelerate solutions for connectivity in various sectors including agriculture,” he says.

In some cases, even when there

are no financial constraints, farmers are still unable to benefit from a solution because of illiteracy and language barriers. Being able to converse with a tech platform in their native language might enable more farmers to access technology. At the World Economic Forum in January this year, Microsoft CEO Satya Nadella spoke about how an Indian farmer was able to access online government services with the help of ChatGPT.

In order to demonstrate the benefits of using drones to Indian farmers, drone maker Garuda Aerospace recently signed an MoU with Tata Chemicals’ subsidiary Rallis India Limited. Rallis will supply pesticides and crop nutrients to Garuda Aerospace, which will use drones to spray these on fields.

Khurana suggests that, “The mantras for agri-tech startups should be around low cost and high efficiency, with absolute ease of use. To align better with the Indian farmers’ mindsets and constraints, the startups should also build a close-looped feedback mechanism where they can ensure that their products are viable, implementable, and affordable for small and marginal farmers.”

The way ahead

According to the India Brand Equity Foundation, India ranks third in terms of agri-tech funding and number of agri-tech startups. By 2025, Indian agri-tech companies are likely to witness investments worth US\$30-35 billion.

The Union Budget for 2023-24 aims to give a big boost to Indian agri-tech startups. Finance Minister Nirmala Sitharaman announced that a new ‘Agriculture Accelerator Fund’

Experts



Rajesh Jalan, Chief Technology Officer, Cropin



Shweta Khurana, Senior Director - APJ, Government Partnerships & Initiatives, Global Government Affairs, Intel

would be set up to encourage agri startups founded by young entrepreneurs in rural areas. “The Fund will aim at bringing innovative and affordable solutions for challenges faced by farmers. It will also bring in modern technologies to transform agricultural practices, increase productivity and profitability,” she said in her Budget Speech.

She also announced plans to build a digital public infrastructure for agriculture, which will, in her words, “... enable inclusive, farmer-centric solutions through relevant information services for crop planning and health, improved access to farm inputs, credit, and insurance, help for crop estimation, market intelligence, and support for growth of agri-tech industry and startups.”

“I foresee that the farming community in India will drive the mass adoption of technology in the coming years, similar to the fintech revolution in the last decade. The ecosystem is working together to democratise agriculture and help the community with solutions addressing their day-to-day requirements, such as healthy crops, increased yield, and managing any risks proactively. The future of smart farming looks bright and promising in India,” Jalan signs off on a positive note. **EFY**